

Securing Intelligent Transportation Systems: A Dual-Framework Approach for Privacy Protection and Cybersecurity Using Generative AI

Muhammad Attique Khan¹, *Member, IEEE*, Areej Alasiry², Mehrez Marzougui³, Isa Bayhan⁴,
Siva Sarana Kuna, G. Siva Nageswara Rao, Shabbab Ali Algamdi⁵, and Haya Aldossary⁶, *Member, IEEE*

Abstract—Integrating Generative AI (GenAI) into Intelligent Transportation Systems (ITS) raises both enormous opportunities and major worries, especially in the areas of privacy and cybersecurity, which are already at the forefront of these developments. Developing and implementing robust security measures to secure sensitive data and address new cyber threats is of utmost importance, especially with the growing dependence on AI technology in transportation networks. This article looks at GenAI and how it may improve ITS intelligence and efficiency while addressing the risks of using it a lot. It delves into the difficulties of protecting AI-driven systems against hostile assaults (AI-MA), particularly emphasizing transportation infrastructure security, intrusion detection, and data privacy. The research stresses the significance of modern encryption methods, real-time monitoring threats, and adaptive security frameworks to ensure ITS are secure and resilient. In addition, it delves into how transportation systems are affected by ever-changing cyber threats, offering proactive security solutions to combat these dangers and strengthen ITS. This paper's overarching goal is to lay out a course of action for integrating GenAI into ITS in a way that strikes a good balance between fostering innovation and ensuring privacy and security via thorough analysis. The proposed AI-MA model achieves a high threat detection accuracy of 96.2%, a privacy protection score of 91.8%, a computational efficiency of 92.9%, a resilience score of 97.8%, and a network reliability ratio of 92.6% compared to other existing models.

Index Terms—GenAI, privacy protection, security protocols, cyber threats, intelligent transportation system.

I. INTRODUCTION

SEVERAL advantages and disadvantages, especially concerning privacy and cybersecurity, come with the growing use of GenAI in ITS [1]. The security risks posed by GenAI are substantial, including data breaches, adversarial assaults, and unlawful system manipulation [2], even though it offers several advantages, such as better traffic optimization, predictive analytics, and timely decision-making. Cyberattacks such as deepfake attacks on traffic monitoring, illegal data access, and AI-driven malware are a major concern for AI-driven ITS [3]. Intelligent and adaptive cybersecurity solutions are necessary because traditional security solutions cannot handle these new threats [4]. Strong encryption and privacy-preserving methods are top priorities since ITS is based on vast quantities of sensitive data, such as data passed between vehicles and everything else, users' locations, and automatic decision-making output [5]. Real-time detection of cyber-attacks is another challenge, with current solutions unable to differentiate between usual system behavior and advanced attacks with artificial intelligence (AI) [6]. The non-traditional and decentralized security mechanisms result from the decentralized environment of modern transport systems [7]. The development of GenAI in ITS without a robust cybersecurity system poses the threat of catastrophic failures, undermined user protection, and deep-rooted system disruptions [4]. To alleviate these issues, this study conceptualizes a two-framework framework for protecting privacy and systems within ITS environments against evolving cyber threats [5]. The solution includes adaptive security mechanisms, time-based threat monitoring, and advanced encryption techniques.

Various techniques have been employed to enhance privacy and cybersecurity in ITS; however, these techniques are not sufficient to handle emerging threats fueled by AI [6]. Security of data transmission within ITS is based on conventional encryption techniques such as Public Key Infrastructure (PKI) and Advanced Encryption Standard (AES) [8]. Real-time processing and quantum computing threats are needed more than they can manage [9]. Most organizations install intrusion detection systems (IDS) to track suspicious network traffic,

Received 22 March 2025; revised 1 May 2025 and 2 June 2025; accepted 16 July 2025. This work was supported by the Deanship of Research and Graduate Studies at King Khalid University through Large Research Project under Grant RGP2/471/46. The Associate Editor for this article was S. Rani. (*Corresponding author: Muhammad Attique Khan.*)

Muhammad Attique Khan is with the Department of Artificial Intelligence, Prince Mohammad Bin Fahd University, Dhahran 34754, Saudi Arabia (e-mail: mkhan3@pmu.edu.sa).

Areej Alasiry and Mehrez Marzougui are with the College of Computer Science, King Khalid University, Abha 61413, Saudi Arabia (e-mail: areej.alasiry@kku.edu.sa; mhrez@kku.edu.sa).

Isa Bayhan is with the Tourist Guiding Department, Bolu Abant Izzet Baysal University at Golkoy, 14030 Bolu, Türkiye (e-mail: isa.bayhan@gmail.com).

Siva Sarana Kuna resides in Holly Springs, NC 27540 USA (e-mail: ssk0315515865@gmail.com).

G. Siva Nageswara Rao is with the Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation, Vaddeswaram, Andhra Pradesh 522302, India (e-mail: sivanags@kluniversity.in).

Shabbab Ali Algamdi is with the Department of Software Engineering, College of Computer Science and Engineering, Prince Sattam Bin Abdulaziz University, Al-Kharj 16278, Saudi Arabia (e-mail: s.algamdi@psau.edu.sa).

Haya Aldossary is with the Department of Computer Science, College of Science and Humanities, Imam Abdulrahman Bin Faisal University, Jubail 31961, Saudi Arabia (e-mail: Healdossary@iau.edu.sa).

Digital Object Identifier 10.1109/TITS.2025.3591007

and they can employ either a signature-based or anomaly-based method [10]. However, these systems are not always effective in detecting malicious AI-driven software; thus, reactions are slow [5]. Research on blockchain technology has been aimed at improving the security of vehicle-to-everything (V2X) communications through more secure data transmission [6]. While blockchain enhances transparency and trust, it suffers from scalability issues and high computational overhead and is not appropriate for application in ITS systems based on real-time data [11]. Decentralized AI-based security protocols are now achievable with the advent of federated learning, reducing the centralized data storage requirement [10]. However, its effectiveness is hampered by challenges such as model poisoning attacks and resource constraints in edge devices [12].

In addition, privacy-preserving techniques try to protect user data, e.g., homomorphic encryption and differential privacy. Still, these techniques often introduce trade-offs between computation efficiency and security. A problem with such approaches is that they cannot switch to address the continuously evolving cyber threats impacting ITS. For resilience, this paper presents a responsive, dual-architecture solution with better encryption, real-time threat monitoring, and self-adjusting cybersecurity protocols. This method attempts to solve this study's drawbacks.

A. Motivation

The proliferation of ITS has transformed modern-day mobility. ITS allows autonomous decision-making, real-time traffic management, and forecasting analytics. Improved ITS performance and capability are potential gains in applying Generative AI (GenAI). However, a serious security and privacy risk lies with this advance. ITS reliability and security are heavily under threat by cyber threats such as AI-based malware, deepfake assaults against monitoring infrastructure, and data leakages by unauthorized access. Centralized security architectures, signature-based intrusion detection, and conventional encryption cannot keep up with the speed of these sophisticated assaults. Consequently, an advanced and adaptable cybersecurity infrastructure capable of detecting and countering attacks driven by artificial intelligence is necessary for ITS privacy protection. An important risk for AI-driven ITS is cyberattacks including malware, illicit data access, and deepfake attacks on traffic monitoring. Because traditional cybersecurity solutions are unable to manage these emerging threats, intelligent and adaptive cybersecurity solutions are required. The approach employs AI-driven intrusion detection systems (IDS) to monitor unusual or malicious behavior and self-adjust security protocols to change encryption and firewall settings in reaction to real-time threats. Improve the resilience of ITS against cyber threats with real-time threat monitoring and fast incident response capabilities. These features help identify assaults early, isolate affected systems, and reduce their effects.

B. Problem Statement

ITS' current security systems have limitations despite AI and cyber security advancements. Despite their effectiveness,

encryption mechanisms like AES and PKI are computationally intensive and non-real-time flexible. False alarms and deteriorated responses are caused by intrusion detection systems not being able to distinguish between malicious and benign AI-generated abnormalities. Blockchain-based security systems guarantee data integrity but are expensive to process and have poor scalability. Provide system reliability, privacy protection, and data integrity to improve the framework. This strategy provides a great way to foster overall transportation security by fusing necessary protection with innovation. Organizations can improve their security posture, manage and lower cybersecurity risks, and stay in compliance with applicable legislation by using security frameworks, which are defined rules and best practices. The decentralized and dynamic nature of ITS is a threat to the ever-evolving cyber world. Public trust is lost, users are compromised, and transportation systems are impaired by cyberattacks when it has no dynamic and intelligent security system.

C. Contribution

To solve the privacy and cybersecurity challenges of ITS, this research proposes a double-framework approach that utilizes Generative AI. Some of the proposed frameworks include:

- Generative AI is utilized to identify anomalies and counter adversarial attacks on ITS networks in real-time.
- Advanced encryption techniques are applied to ensure data integrity while allowing computing efficiency.
- A security model based on AI-MA should be created to learn how to counter new threats as they emerge.
- Secure data sharing with minimal risk of exposure through federated learning and differentiated privacy.

The research paper's structure is laid out in this section, including the following: Section II focuses on integrating security measures into ITS. The Section III of this dissertation delves into the topic of AI-driven systems and how they protect against malicious attacks (AI-MA). Section IV provides an in-depth examination, a review of related approaches, and an interpretation of the results and their significance. This study's findings are discussed in detail in Section V.

II. LITERATURE SURVEY

The quick advances in ITS have raised critical cybersecurity issues, and powerful defense systems are needed to cope with emerging threats. Various cybersecurity approaches have been developed to enhance ITS security, each with advantages and limitations.

A. Cybersecurity Measures in ITS: Strengths and Limitations of Existing Approaches

The suggested Multicriteria Decision-Making (MCDM)-Based Cybersecurity Metrics Evaluation approach by Bhol et al. [13] employs five critical metrics to quantify cybersecurity strength. Permits the methodical evaluation of security efficacy—Trickiness in determining the best weights for criteria. Improving decision-making using MCDM necessitates careful metric selection for reliable cybersecurity evaluation.

The Cybersecurity-Integrated Machine Learning (CIML) Framework, developed by Wazid et al. [14], improves security and can identify zero-day attacks. Its benefits lower the need for human involvement. Negative aspects adversely affected by hostile assaults. Inference Strong protections against ever-changing threats are necessary for CIML to boost cybersecurity efficiency.

B. AI-Driven Threats in ITS: Challenges in Securing Intelligent Systems

Chaudhary suggested Cybersecurity Awareness Evaluation Metrics (CSA-EM) [1] to assess cybersecurity awareness programs fully. Ensures complete, reproducible measurement. No standards for measurement exist. While CSA-EM enhances program performance, it must be developed further before it can be applied more extensively.

Machine learning-based cybersecurity risk detection (ML-CRD) is a method that analyzes cyber threats, as suggested by Shaukat et al. [2]. ML-CRD improves cybersecurity, though it requires optimization to function optimally.

C. Advancing ITS Security: The Role of Generative AI in Privacy Protection and Cybersecurity

Pollini proposed a Human Factors-Based Cybersecurity Framework (HF-CF) [3], incorporating user behavior into CIS. HF-CF enhances cybersecurity by finding a middle ground between technological and people-focused strategies.

Cyber hazards can be measured using the Decision-Analysis-Based Cybersecurity Framework (DAB-CF) proposed by Ganin et al. [4]. Decisions are made subjectively. DAB-CF improves the management of cybersecurity risks in an organized and transparent manner.

A comparison of cybersecurity measures with existing studies is made in Table I.

Although existing methodologies adequately illuminate the means of protecting ITS, they possess limitations such as high computational intensity, conflicts in compliance, and dependence on personal judgment. The AI-driven systems Against Malicious Attacks (AI-MA) approach outshines other approaches using advanced generative AI technologies for adaptive threat recognition and reaction. AI-MA enhances cybersecurity effectiveness by stopping threats in real-time with reduced human intervention. In addition, it enables methods that protect privacy, such as synthetic data production, which lessens the need for raw personal data. Negative aspects include, but are not limited to, the following: increased complexity of systems, which increases the attack surface; the possibility of adversarial attacks generated by GenAI that could fool perception and decision-making systems; and an amplified risk to privacy if synthetic outputs unintentionally reconstruct sensitive information. Using GenAI, bad actors can create believable phishing, spoofing, or data poisoning attacks against ITS systems.

III. PROPOSED METHOD

Though Generative AI (GenAI) in ITS has advantages, data privacy and cybersecurity are at risk. Transportation systems

TABLE I
COMPARISON OF CYBERSECURITY MEASURES

Author(s)	Proposed Method	Advantages	Disadvantages
Bhol, S. G. et al.	Multicriteria Decision-Making (MCDM)-Based Cybersecurity Metrics Evaluation	Enables systematic evaluation of cybersecurity strength	Difficulty in determining optimal criteria weights
Wazid et al.	Cybersecurity-Integrated Machine Learning (CIML) Framework	Reduces human intervention, detects zero-day attacks	Susceptible to adversarial attacks
Chaudhary, S. et al.	Cybersecurity Awareness Evaluation Metrics (CSA-EM)	Provides comprehensive and reproducible measurement	Lacks standardized measurement criteria
K. Shaukat et al.	Machine Learning-Based Cybersecurity Risk Detection (ML-CRD)	Enhances accuracy of threat detection	Requires high computational resources
Pollini, A. et al.	Human Factors-Based Cybersecurity Framework (HF-CF)	Strengthens cybersecurity culture, integrates user behavior	Potential conflicts with compliance policies
Ganin, A. A. et al.	Decision-Analysis-Based Cybersecurity Framework (DAB-CF)	Integrates risk factors systematically	Decision-making remains subjective

depend on AI-based solutions to control growing cyber risks, and it is crucial to apply thorough security policies. Generative AI (GenAI) can potentially improve traffic estimates on both large and small scales via its capacity to analyze and forecast massive volumes of real-time data. To improve dynamic traffic management and eliminate bottlenecks, GenAI can examine citywide traffic patterns, weather, and events on a macro scale, allowing it to forecast congestion and optimize traffic flow over vast regions.

Fig. 1 shows the dual framework security concept to enhance protection against ITS cyberattacks. Privacy protection and cybersecurity are the two main components of the framework, which rests under one security layer. Privacy is protected through federated learning, encryption, and data anonymization. These three methods protect sensitive transportation information from being released to the public [16]. An approach ensures that GenAI, which is utilized in ITS, eliminates information abuse while keeping data privacy and cybersecurity integrity in transportation systems [17].

$$\partial_p f = Cx \left[a + nr'' \right] * Vx \left[a - mj'' \right] + yr \left[s - ki' \right] \quad (1)$$

Including adaptive security parameters including real-time threat mitigation ($Vx \left[a - mj'' \right]$), safeguarding measures ($\partial_p f$), and detection of intrusion signals ($Cx \left[a + nr'' \right]$), Equation (1) illustrates the dynamic interaction between

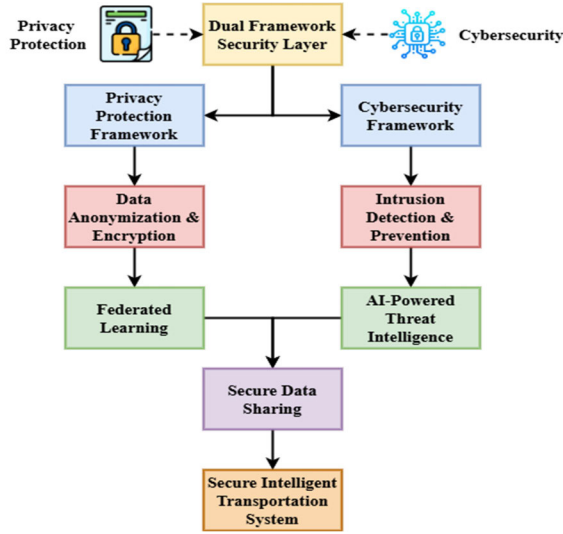


Fig. 1. Dual framework security model for ITS.

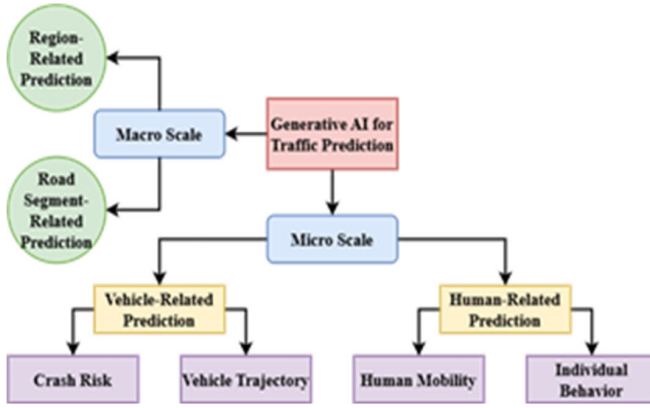


Fig. 2. Generative AI-Based Traffic Prediction Model.

confidentiality (yr) and security issues ($[s - ki']$) in ITS.

$$R_{fg} = tr[a + nr''] + ew[q + mz'] * v[a - hr'] \quad (2)$$

Equation (2) describes the resilience measure (R_{fg}) of ITS to assess system robustness by combining real-time threat recognition ($tr[a + nr'']$), data encryption weight ($v[a - hr']$), and adaptive protection response ($ew[q + mz']$).

$$\partial_{ar} = bf[r + xk''] - V[a - mju''] * h[e - tu'] \quad (3)$$

By balancing threat effect ($bf[r + xk'']$), privacy-preserving data encryption ($V[a - mju'']$), and changing safety responses ($h[e - tu']$), Equation (3) describes the adaptive risk component (∂_{ar}) in ITS. The ITS coordinator of communication flows is the nerve center that controls the flow of information between cars and traffic control systems, allowing for a precise and rapid reaction to any dangers that can be recognized. Vehicle and roadside sensors instantly notify a centralized traffic management system of any danger, such as an accident or blockage.

Fig. 2 illustrates how GenAI can assist in estimating traffic with macro and micro-scale studies. Macroscale forecasting highlights region road segment-oriented traffic conditions favoring prominent transportation planning and congestion alleviation. GenAI offers detailed micro-scale analysis concerning human- and vehicle-oriented behavior. Vehicle-related prediction enhances traffic flow management and road safety by analyzing vehicle trajectory and accident risk. Human forecasting is the backbone of pedestrian safety and adaptive traffic management and emphasizes individual mobility and behavior patterns. By employing actual-time, fact-based insights based on aggregating these predictive factors, the method proposed improves ITS [18]. It diminishes traffic congestion and enhances overall transportation efficiency by enabling proactive decision-making. A comprehensive framework for maximizing urban mobility and safeguarding ITS against emerging threats is offered by macro and micro-scale traffic forecasting. The demands of the populace must be met by a comprehensive mobility plan that prioritizes public and non-motorized transportation above private automobiles as a means of successfully combining land use and transportation planning.

$$\tau_{ft} = Ev[a + mk''] * cs[a + kd''] + y[e - wq'] \quad (4)$$

Incorporating safeguarding validation ($Ev[a + mk'']$), cybersecurity power ($y[e - wq']$), and flexible threat mitigation ($cs[a + kd'']$) Equation (4) establishes the security factor of confidence (τ_{ft}) in ITS. This suggested a dual-framework strategy by measuring the contribution to improving real-time threat mitigating and secure communications.

$$\tau_{px} = Cx[a + nr''] * ew[a + br''] + t[r - bw'] \quad (5)$$

Integrating safeguarding weight (τ_{px}), network assessment of risk ($Cx[a + nr'']$), and threat impact mitigating ($ew[a + br'']$), Equation (5) reflects the privacy endurance factor ($t[r - bw']$) in ITS. This quantifies how generative AI improves data privacy while mitigating real-time cyber risks.

$$\tau_{aqm} = vx[a + slo''] * vd[s + ny''] + cx[s - ui'] \quad (6)$$

Integrating the loads of system optimization (τ_{aqm}), vulnerability identification ($vx[a + slo'']$), and illegal access countermeasures ($vd[s + ny'']$), Equation (6) describes the adaptive privacy factor ($cx[s - ui']$) in ITS. The Equation provides dynamic threat response and system resilience, guaranteeing safety and confidentiality protection in ITS. ITS has integrated cutting-edge technology to improve transportation safety, efficiency, and sustainability, and their broad adoption has transformed contemporary transportation. ITS optimizes traffic flow, reduces congestion, and minimizes delays by using real-time data collected from sensors, cameras, and GPS systems to manage dynamic traffic. There has been a marked improvement in traffic safety thanks to adaptive traffic signal systems that react in real-time to traffic circumstances.

Algorithm 1 AI-Powered Intrusion Detection System (AI-IDS) for ITS

```

def ai_intrusion_detection(anomaly_score, threshold):
    if anomaly_score > threshold:
        print("ALERT: Potential intrusion detected! Initiating
countermeasures...")
        block_suspicious_ip()
        alert_security_team()
    else:
        print("System secure. No intrusion detected.")
def block_suspicious_ip():
    print("Blocking the suspicious IP address...")
def alert_security_team():
    print("Notifying security team for further investigation...")
anomaly_score = 0.85
threshold = 0.75
ai_intrusion_detection(anomaly_score, threshold)

```

This Algorithm 1 monitors network traffic using an AI-derived anomaly score. If the score exceeds a set threshold, it triggers alerts, blocks suspicious IP addresses, and notifies the security team. Otherwise, it indicates a secure system. This proactive approach effectively enhances real-time threat detection and rapid response in intelligent transportation networks. By lowering system availability, tampering with traffic control algorithms, and exposing crucial infrastructure to manipulation or disruption, evolving cyber threats like APTs, coordinated botnet attacks, data spoofing, and AI-generated adversarial examples directly affect the reliability and security of transportation networks. System performance metrics, encrypted communication logs, and intrusion attempts are used to evaluate the efficacy of security. GenAI for threat detection, privacy protection, and resilience assessment is trained and validated using this dataset. Real-time traffic data, cyber threat logs, and ITS anomaly detection records are all included in the analytical dataset. The ever-changing nature of these dangers makes it more likely that linked ITS components fail one after the other, damages trust models, and makes sensor data less reliable. These measures include using blockchain-based identity management for decentralized authentication, monitoring distributed edge nodes without centralized vulnerabilities through federated learning models for real-time anomaly detection, applying moving target defense (MTD) techniques to randomize system configurations and limit attack surface exposure continuously, and enforcing dynamic encryption key rotation synchronized with network state changes to minimize the window of exploitation.

Fig. 3 identifies the safety issues created by GenAI entering ITS and the recommended actions to mitigate these risks [19]. GenAI is primarily accountable for traffic forecasting and autonomous vehicle driving, with increased efficiency and decision-making. Proactive decision-making is made possible, which reduces traffic congestion and improves overall transportation efficiency. Traffic forecasting at both the macro and micro scales offer a comprehensive framework for maximizing urban mobility and safeguarding ITS against emerging threats. Its usage does have its downfalls, as autonomous vehicle systems and predictive models can be leveraged in adversarial attacks under the control of AI. These attacks can compromise

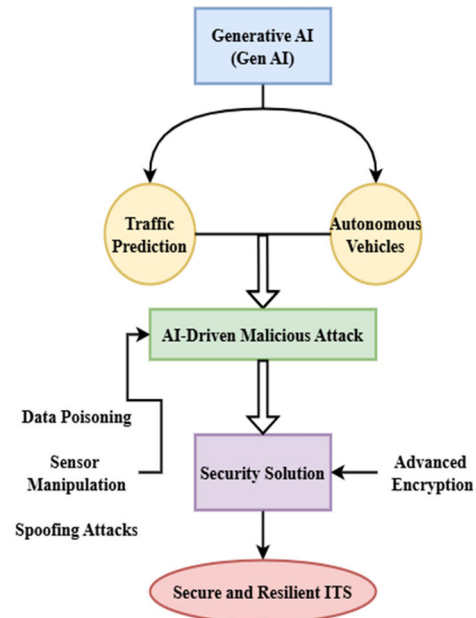


Fig. 3. Security challenges and solutions in GenAI-Driven ITS.

the reliability of ITS using traffic management methods, illegal control, or data leakage. By utilizing advanced security protocols employed by the system, such hazards are minimized through strong encryption techniques, machine learning-based anomaly detection, and real-time attack detection [20].

$$\tau_v x = kt [d + nx''] - j [a - xk'] * vx [a + sp''] \quad (7)$$

Incorporating the network of things threat detection ($\tau_v x$) policy on security enforcement ($j [a - xk']$), and the method protection measures ($kt [d + nx'']$), Equation (7) establishes the vulnerability mitigating factor ($vx [a + sp'']$) in ITS. This suggested dual-framework strategy as it quantifies how generative AI helps to lower data privacy violations.

$$\partial_v s = vx [a + sx''] * re [w + nji''] + vd'' \quad (8)$$

Equation (8) integrates system resilience ($vx [a + sx'']$), instantaneous data encryption (vd''), along with vulnerability assessment ($re [w + nji'']$), thereby modeling the security versatility factor ($\partial_v s$) in ITS. This is consistent with the suggested dual-framework strategy: generative AI improves adaptive security mechanisms to fight developing cyber threats.

$$\forall_z q = tr [k - sjy''] + re [w + naw''] * n [o - p'] \quad (9)$$

Equation (9) integrates threat reduction ($\forall_z q$), immediate encryption ($tr [k - sjy'']$), and network anomaly identification ($re [w + naw'']$), thereby representing the universal security positive reinforcement take $n [o - p']$ in ITS. This suggested dual-framework approach by generative AI improves proactive security systems to reduce cyber threats and defend privacy.

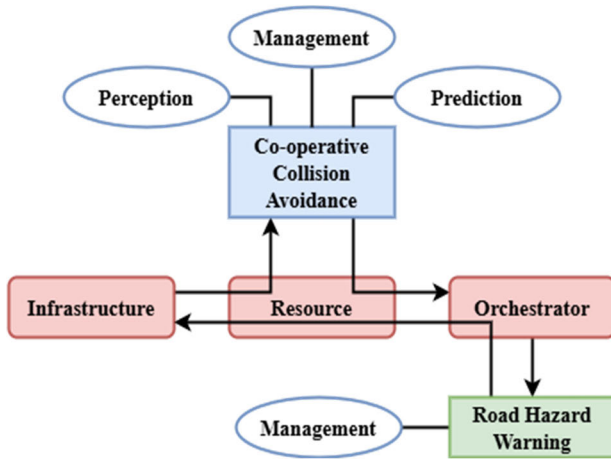


Fig. 4. Cooperative Collision Avoidance in ITS.

Fig. 4 shows a cooperative collision avoidance strategy to improve traffic safety with ITS technologies. The approach combines economy, management, technology, and constituents, making it possible to make real-time decisions with minimal mistakes [21]. Using ITS technologies, an avoidance strategy can increase traffic safety. The strategy integrates constituents, technology, management, and economy to enable real-time decision-making with few errors. To detect and reduce potential problems, the coordinated strategy makes use of data from several sources. Sensors and the required degree of communication are provided by infrastructure components. However, the optimal and efficient management of computational and network resources assures perfect data processing. Data processing perfection is ensured by the effective and efficient use of network and computational resources. The communication flow coordinator makes it possible for cars and traffic control systems to react precisely to hazards that are identified. Furthermore, raising awareness means alerting autonomous cars to potential dangers and obstructions on the road. In addition, increasing awareness indicates notifying self-driving vehicles of possible risks and road obstacles. The combined use of GenAI and cooperative collision avoidance systems creates powerful and safe ITS that reduces road accidents and enhances transportation efficiency through predictive analytics and adaptive response technologies.

$$\partial_l c x = [a + jv''] * yt[s + nmk''] + r[s - az'] \quad (10)$$

By including adaptive threat response ($\partial_l c x$), system resilience ($[a + jv'']$), and anomaly mitigating ($yt[s + nmk'']$), Equation (10) describes the latent cyber-risk factor ($r[s - az']$) in ITS. This generative AI improves cybersecurity protections while preserving privacy protection.

$$nz! = F[nS + \sigma\epsilon''] - \pi\beta[\tau\sigma + yrw''] * \mu\vartheta c'' \quad (11)$$

Incorporating method sensitivity ($nz!$), security data encryption ($F[nS + \sigma\epsilon'']$), and response to threats ($\pi\beta[\tau\sigma + yrw'']$), Equation (11) predicts the advanced cybersecurity risk factor ($\mu\vartheta c''$). Integrating these factors

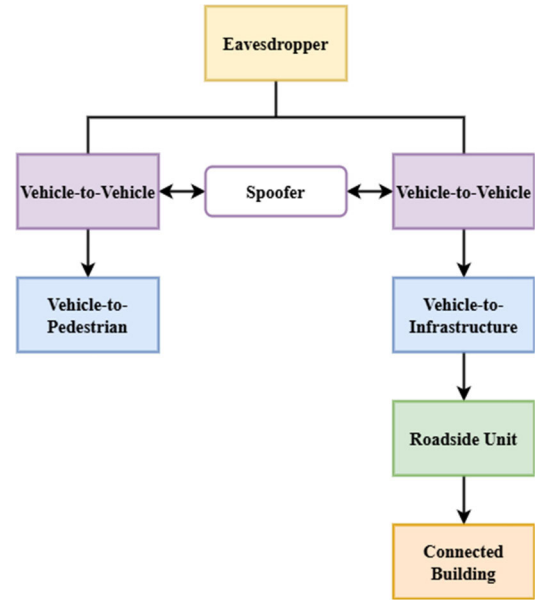


Fig. 5. Block diagram secure communication in intelligent transportation systems.

helps the Equation enable a proactive, powered by AI security policy to guarantee robust data protection.

$$\tau_l c = tr[s + nc''] * xz[a + br''] + re[s - ja'] \quad (12)$$

Integrating risk mitigation ($\tau_l c$), system fragility ($tr[s + nc'']$), and encryption ($xz[a + br'']$), Equation (12) establishes the cyber endurance factor ($re[s - ja']$) in ITS. This is consistent with the suggested dual-framework strategy: AI preserves privacy while enhancing security mechanisms against new risks.

Fig. 5 focuses on protecting ITS, thus filling gaps in the communication networks. Cyberattacks that include spoofing, eavesdropping, jamming, compromising Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Pedestrian (V2P), and Vehicle-to-Network (V2N). The figure illustrates probable attack sites, such as a spoofer altering V2V communication, an eavesdropper assembling V2I information, and a jammer disrupting railway signals [22]. A scalable security architecture and robust encryption ensure secure data transport and defense against new threats. To increase our systems' resistance to attacks, we use AI-based anomaly detection and secure authentication methods. These security methods provide data integrity, privacy protection, and system reliability in GenAI-based ITS, strengthening the architecture.

Secure authentication techniques and AI-based anomaly detection are employed to boost the robustness of our systems to attacks [23]. In GenAI-based ITS, these security techniques deliver data integrity, privacy safeguarding, and system dependability, thus enhancing the framework. Such an approach offers an excellent way of combining required security with creativity to promote overall transport security.

$$\tau\mu_x a = jr[v\pi + ba''] - \sigma\vartheta'[\varepsilon\gamma + baw''] + \pi\sigma\tau'' \quad (13)$$

Algorithm 2 Secure AI-Driven Communication in ITS

```

def secure_data_transmission(is_encrypted, sender_verified):
    if is_encrypted:
        if sender_verified:
            print("Data transmission secure. Proceeding with
communication.")
        else:
            print("WARNING: Unverified sender! Blocking
communication.")
            block_unverified_sender()
        else:
            print("ERROR: Data not encrypted! Enforcing encryption
protocols...")
            enforce_encryption()
    def block_unverified_sender():
        print("Blocking data packets from unverified sender...")
    def enforce_encryption():
        print("Applying end-to
        -end encryption to secure transmission...")
    is_encrypted = True
    sender_verified = False
    secure_data_transmission(is_encrypted, sender_verified)

```

integrating real-time threat identification ($\sigma \vartheta' [\varepsilon \gamma + baw'']$), system encryption ($j r [v\pi + ba'']$), and adaptive risk mitigation ($\tau \mu_x a$), Equation (13) describes the advanced protection adaptability take ($\pi \sigma \tau''$) in ITS. This is consistent with the suggested dual-framework strategy as it underlines how generative AI improves confidentiality adaptive actions.

$$\forall_c q = tr [\vartheta + baw''] * tr [\theta \delta + bnr''] + \mu \rho'' \quad (14)$$

By combining dynamic threat identification ($tr [\theta \delta + bnr'']$), adaptive risk management ($tr [\vartheta + baw'']$), and anomaly detection ($\forall_c q$), Equation (14) describes the whole security factor ($\mu \rho''$) in ITS. This suggested a dual-framework strategy by measuring how generative AI improves ITS proactive protection systems monitoring.

$$R_z x = kp [\aleph s + yr''] * C [a - nr''] + \delta \gamma t'' * vx'' \quad (15)$$

Integrating the identification of threats ($R_z x$), system safeguarding ($k p [\aleph s + yr'']$), and risk-mitigating ($C [a - nr'']$), Equation (15) describes the security endurance factor ($\delta \gamma t'' * vx''$) in ITS. The Equation provides adaptive defenses, strengthening AI-driven transit systems' cybersecurity and resilience [24].

Algorithm 2 secures AI-driven communication in ITS by verifying encryption and sender authenticity. This function verifies the sender's identity and determines if the data is encrypted. To safeguard AI-driven ITS, this article suggests a security architecture that includes adaptive security mechanisms, better encryption, and real-time threat monitoring [25]. By utilizing the system's sophisticated security procedures, such risks are reduced by real-time attack detection, machine learning-based anomaly detection, and robust encryption approaches. By protecting transport systems against cybercrime, these protocols secure data transfer and ITS operations.

TABLE II
SIMULATION AND ENVIRONMENT TABLE

Parameter	Description
Simulation Tool	MATLAB, Python (TensorFlow, PyTorch), NS-3
Computational Setup	High-performance computing (HPC) cluster, GPU acceleration
Network Model	Intelligent Transportation System (ITS) simulation
Security Framework	Generative AI (GenAI)-based Intrusion Detection System (IDS), Blockchain-based authentication
Data Source	Real-time traffic logs, cyber threat databases, intrusion logs
Evaluation Metrics	Detection accuracy, false positive rate, computational latency, resilience score
Attack Scenarios	Adversarial AI attacks, DDoS, Man-in-the-Middle (MITM), Data Breaches
Encryption Standards	AES-256, Lightweight cryptographic algorithms
Simulation Duration	24-hour real-time analysis with periodic updates

By incorporating advanced encryption techniques, real-time threat monitoring, and adaptive security frameworks, ITS is strengthened in terms of safety and resilience. Lightweight elliptic curve cryptography and post-quantum algorithms are two examples of advanced encryption methods that safeguard vehicle-to-infrastructure (V2I) and vehicle-to-vehicle (V2V) communications against manipulation and eavesdropping while preserving low latency. AI-based intrusion detection systems can detect and isolate real-time abnormalities by analyzing sensor data, network traffic, and behavioral patterns, preventing cascading failures. Adaptive security frameworks adapt defensive mechanisms depending on changing threat landscapes by incorporating continuous learning, behavior analytics, and context-aware risk assessment.

IV. RESULTS AND DISCUSSION

Security and performance in ITS require cyber threat mitigation, computational efficiency, resilience, reliability, and privacy protection. GenAI enhances real-time threat detection, anomaly detection, and secure data sharing.

The analytical dataset [15] includes real-time traffic data, cyber threat logs, and ITS anomaly detection records. Security effectiveness is assessed using encrypted communication records, intrusion attempts, and system performance indicators. The dataset trains and validates GenAI for threat detection, privacy protection, and resilience evaluation.

In the above table II, the simulation environment uses NS-3, Python (TensorFlow, PyTorch), and MATLAB on a GPU-accelerated HPC cluster. It tests the safety of ITS against a variety of cyber threats utilizing methods such as AES-256 encryption, blockchain authentication, and intrusion detection systems based on GenAI. To achieve the necessary level of connectivity and sensor integration in ITS, infrastructure components like Roadside Units (RSUs), On-Board Units (OBUs), centralized Traffic Management Centers (TMCs), edge computing nodes, and Vehicle-to-Everything (V2X) communication modules are utilized. OBUs handle vehicle-specific

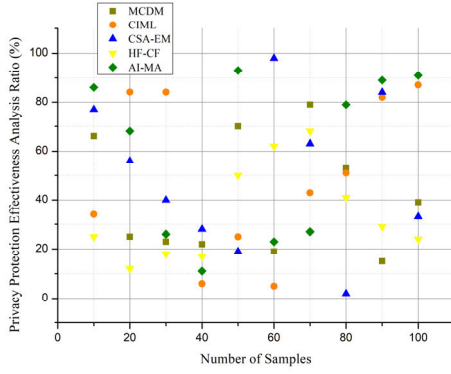


Fig. 6. Privacy protection effectiveness analysis.

TABLE III
CYBER THREATS IN ITS (2018–2025)

Year	Data Breaches	Malware Attacks	AI-Powered Attacks	Ransomware Cases
2018	50	80	5	20
2019	70	120	10	35
2020	110	150	20	50
2021	150	200	40	80
2022	200	250	70	120
2023	260	320	110	180
2024	310	400	150	250
2025*	400	500	200	300

sensing and local decision-making, while RSUs provide vehicle-to-infrastructure connectivity and collect environmental data. Edge nodes reduce latency for real-time answers by processing data closer to the source.

From the above fig. 6, ITS with GenAI enhances efficiency at the expense of privacy. Real-time threat monitoring, strong encryption, and adaptive security models maintain privacy. AI-MA offers real-time anomaly detection and response and works well in identifying emerging cyber threats. Model updates are necessary because of adversarial attacks. IDS and GenAI identify unwanted attempts at access to enhance cybersecurity resiliency.

$$\partial_{fr} = \partial \forall [a + br''] * [\alpha + vre''] + \forall Xa [\pi \rho''] \quad (16)$$

Integrating adaptive precautions ($\partial \forall [a + br'']$), threat mitigation ($[\alpha + vre'']$), and safeguarding privacy encryption (∂_{fr}), Equation (16) describes the dynamic risk component ($+\forall Xa [\pi \rho'']$) in ITS. Data privacy and cybersecurity are actively controlled, strengthening the security regarding AI-driven analysis of privacy protection effectiveness.

A. Cyber Threat Incidents in ITS Over Time

Objective: Illustrates the increasing frequency of cyber threats targeting ITS from 2018 to 2025 (projected).

Fig. 6(a) and Table III show the rise in cyber threats targeting ITS from 2018 to 2025. AI-powered attacks and

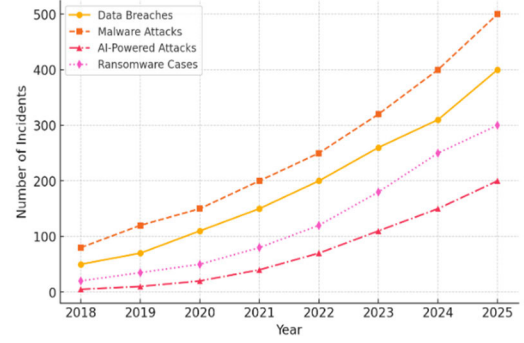


Fig. 7. (a) Cyber threat incidents in its over time.

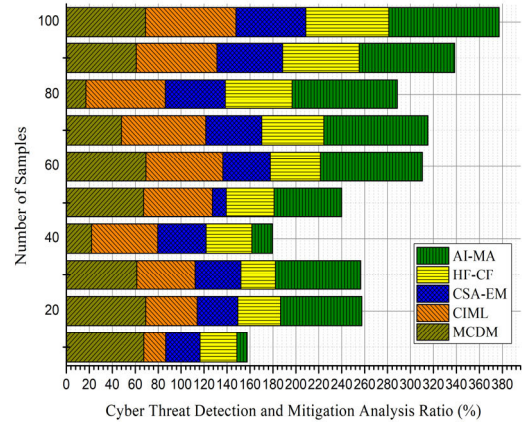


Fig. 8. Cyber threat detection and mitigation analysis.

ransomware cases have increased significantly, emphasizing the growing risks of evolving cyber threats. The data highlights the urgent need for improved cybersecurity measures in ITS.

AI-powered security systems are essential for defending against changing cyberattacks in ITS. GenAI-based IDS enhance anomaly detection through real-time identification of malicious patterns. From the above fig. 7, predictive threat analysis is enhanced by AI-MA algorithms, while blockchain-based authentication secures data. MFA and end-to-end encryption minimize illegal access.

$$\aleph_{ze} = \pi \theta [\omega + ut''] * Vx [\mu + baq''] + \sigma \omega'' \quad (17)$$

Integrating encryption measures ($\aleph_{ze}$), resilient system resilience ($\pi \theta [\omega + ut'']$), along with real-time detection of threats ($Vx [\mu + baq'']$), Equation (17) describes the security improving factor ($\sigma \omega''$) in ITS. This dual-framework strategy uses generative AI to improve adaptive analysis of cyber threat detection and mitigation.

B. Efficiency of Intrusion Detection Systems (IDS) in ITS

Objective: Compares detection rates of different security approaches, including Traditional IDS, AI-powered IDS, and GenAI-Enhanced IDS.

Fig. 7(a) and Table IV compares Traditional IDS, AI-Powered IDS, and GenAI-Enhanced IDS. GenAI-based IDS

TABLE IV
IDS EFFICIENCY COMPARISON (%)

Security Approach	Detection Accuracy	False Positive Rate	Response Time (ms)
Traditional IDS	75	12	250
AI-Powered IDS	85	8	180
GenAI-Enhanced IDS	95	3	100

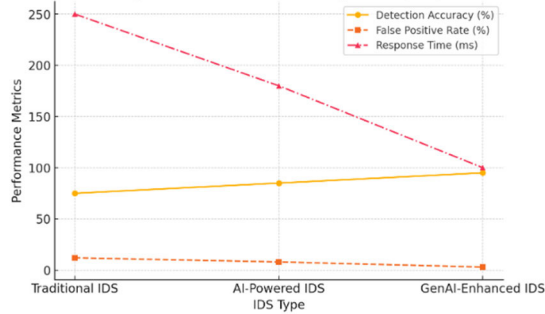


Fig. 9. (a) Efficiency of intrusion detection systems (IDS) in ITS.

outperforms others, achieving the highest detection accuracy (95%) with the lowest false positive rate (3%) and the fastest response time (100ms). The results emphasize integrating AI-driven security solutions to protect ITS from cyber threats.

One way to make ITS more resilient is to use the responsive dual-architecture method, which combines two important levels of security. Secure data transmission among cars, infrastructure, and control centers is the primary goal of the first layer, which employs sophisticated cryptographic algorithms like elliptic curve or quantum-resistant encryption. Because of this, it can be certain that your critical data will stay private and untouchable. As a second layer, self-adjusting Sensitive data must be protected using a multi-layered security strategy that includes technological, procedural, and physical safeguards. A defined cybersecurity policy, frequent backups, strong access controls, data encryption, and security awareness training are all part of this. Security procedures apply safety operations instructions in conjunction with security policies, regulations, and recommendations. cybersecurity protocols dynamically alter security measures in response to changing threat environments. These procedures enhance the system's responsiveness to novel, unanticipated dangers using machine learning models that study assault trends and make real-time adjustments to security systems.

From the above Fig. 8, ITS security demands computational power for real-time threat detection and mitigation. GenAI streamlines pattern recognition and anomaly detection to minimize false positives and accelerate calculation. Edge computing minimizes latency by processing near data sources, increasing reaction times. Lightweight encryption methods balance security and processing power, ensuring seamless ITS operations. AI-MA models need effective resource management and hardware acceleration because they have high computational requirements. Algorithmic

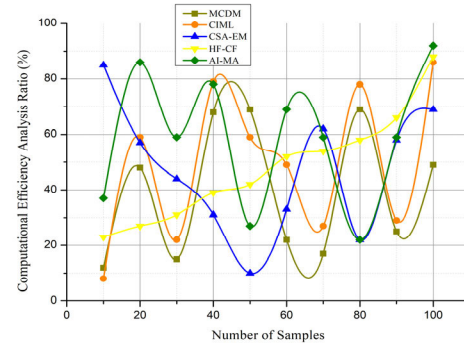


Fig. 10. Computational efficiency.

complexity and parallel processing enhance performance, making ITS security frameworks efficient and scalable without compromising system responsiveness.

$$a_4\tau = [\pi + vr''] * ew[\epsilon + naw''] + ut[\pi - nc'] \quad (18)$$

Integrating threat mitigation ($a_4\tau$), encryption weight ($[\pi + vr'']$), and adaptive mitigating measures ($ew[\epsilon + naw'']$), Equation (18) describes the general security factor ($ut[\pi - nc']$) in ITS. These components of AI-enhanced security solutions are designed to provide strong safeguards and resilience in analyzing computational efficiency. Various measures have enhanced ITS cybersecurity and privacy. Data confidentiality is ensured by using encryption methods such as lightweight cryptography, which allows for safe connection between infrastructure and cars. Blockchain technology is used for transparent transaction records and decentralized authentication to lessen the impact of potential failure points. Because sensitive information is never shared, federated learning allows dispersed data processing for anomaly detection while maintaining privacy. Intrusion detection systems (IDS) powered by AI constantly watch for any signs of intrusion, while access control techniques like role-based authentication ensure that only authorized users can access the system. To further ensure the security of personally identifiable information while data is being analyzed, models are trained, data is anonymized, and synthetic data is generated.

From the above fig. 9, the resilience of ITS against cyberattacks is assessed. GenAI improves resilience by learning from changing attack patterns and mitigating threats. AI-MA-driven anomaly detection and blockchain data integrity layers strengthen the system. Enhancing incident response tactics and self-healing mechanisms can boost ITS resilience, ensuring security and operational continuity in dynamic transportation contexts.

$$Ts = [\alpha + bx''] + \forall z[\partial + yre''] * \sigma\epsilon\gamma[a - s'] \quad (19)$$

Integrating the response to threats (Ts), adaptive mitigating ($[\alpha + bx'']$), and system resilience ($+\forall z[\partial + yre'']$), Equation (19) describes the security protection factor ($\sigma\epsilon\gamma[a - s']$) in ITS. This suggested a dual-framework strategy using generative AI to improve resilience analysis.

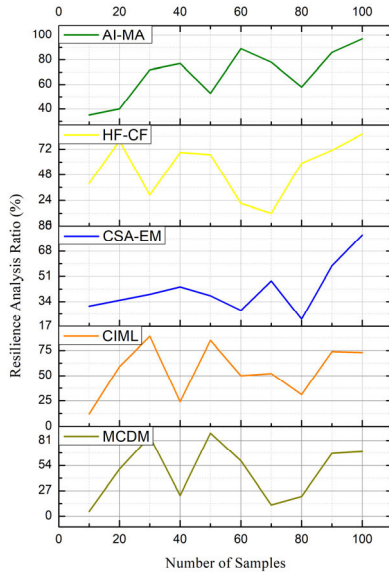


Fig. 11. Resilience analysis.

TABLE V
ENCRYPTION EFFECTIVENESS METRICS

Encryption Method	Processing Overhead (%)	Security Score (/100)	Data Breach Reduction (%)
AES-256	10	85	70
Homomorphic Encryption	25	95	85
Blockchain-Based	15	98	90

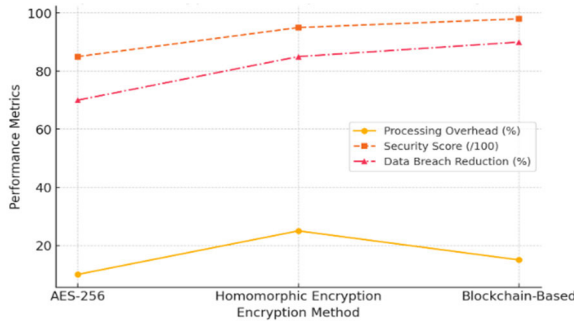


Fig. 12. (a) Encryption techniques on data privacy in ITS.

C. Impact of Encryption Techniques on Data Privacy in ITS

Objective: Evaluate the effectiveness of AES-256, Homomorphic Encryption (HE), and Blockchain-based Encryption in securing ITS communications.

Fig. 9(a) and Table V evaluate AES-256, Homomorphic Encryption, and Blockchain-Based Encryption in securing ITS data. Blockchain-based encryption provides the highest security (98/100) and data breach reduction (90%) with moderate processing overhead (15%). The results suggest that advanced encryption methods enhance data privacy while minimizing computational costs in ITS environments. Modern

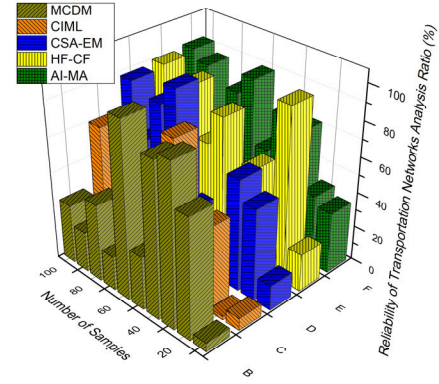


Fig. 13. Reliability of transportation networks.

TABLE VI
COMPARISON OF EXISTING WITH PROPOSED METHOD

Parameter	Existing Methods	Proposed Method AI-MA
Threat Detection Accuracy (%)	65-85%	96.2%
Privacy Protection Score	70-80%	91.8%
Computational Efficiency	50-80%	92.9%
Resilience Score	62-75%	97.8%
Network Reliability (%)	55-80%	92.6%

transportation networks include unconventional and decentralized security mechanisms, such as blockchain-based identity management, allowing for trustworthy peer-to-peer identification independent of any governing body.

From fig. 10, transportation networks must be reliable against cyberattacks for uninterrupted operations. GenAI-driven security frameworks improve reliability by providing predictive threat analysis, real-time anomaly detection, and automated incident response. Secure data transfer and adaptive encryption decrease risks and minimize system disturbances. However, network congestion, latency, and computational resource requirements are concerns. Network resilience is improved via redundancy protocols, blockchain for secure transactions, and self-healing techniques. Strong cybersecurity safeguards ITS from dangers and maintains trust and efficiency.

$$jx[a - r'] = \Delta m [\forall' + te] * vx [\mu a + tw''] \quad (20)$$

Equation (20) integrates the weaknesses of the system (vx), threat growth ($= \Delta m [\forall' + te]$), and adaptive defensive mechanisms ($jx[a - r']$), thereby modeling the security answer factor ($[\mu a + tw'']$). This strategy is based on cybersecurity defensive flexibility in analyzing the reliability of transportation networks.

In Table VI, this comparison shows how the AI-MA security framework is far better than the old techniques in terms of security, efficiency, and resilience.

Findings indicate that AI-MA security frameworks based on GenAI augment ITS security by detecting real-time anomalies, predicting threats, and responding automatically. Privacy,

system adaptation, and computational demand continue to be significant barriers. Edge computing and parallel processing enhance efficiency, whereas self-healing mechanisms and redundancy techniques enhance robustness. Ensuring trust, reliability, and uninterrupted ITS functionality in evolving contexts calls for safe, scalable, and efficient cybersecurity solutions.

Data privacy, effective intrusion detection, and secure communication are the biggest challenges when protecting AI-driven transportation infrastructure systems against malicious assaults (AI-MA). Data aggregation sites are appealing targets for inference and extraction attacks, leading to privacy concerns. These points gather and analyze sensitive real-time information, including vehicle positions, user identities, and behavioral patterns. As malicious actors use intricate evasion strategies, adversarial instances, and zero-day attacks to exploit weaknesses in AI models, intrusion detection systems are becoming more difficult to detect.

Using Generative AI (GenAI) in ITS systems can lead to problems like adversarial input generation, model drift from using synthetic data for training, vulnerability to injection attacks, and more opportunities for reverse engineering to exploit the models. An adversary with malicious intent can use GenAI-driven decision models to cause erroneous sensor data interpretation, dangerous routing, or misclassify vehicle actions. If synthetic data distributions differ from real-world traffic patterns, model drift might negatively affect the operational effectiveness of continuous learning processes driven by GenAI. Hackers can disrupt traffic flow optimization and incident prediction by manipulating real-time GenAI results via quick injection.

First, transportation systems must adhere to several national and international regulations about data privacy, cybersecurity, and traffic management to fulfill regulatory requirements. Data processing, encryption, and accountability are heavily regulated areas; as a result, system changes can be delayed, or AI models can become less flexible. The swift implementation of cutting-edge security measures can be slowed down by the time and effort needed to comply with these rules. Secondly, there is yet another major obstacle in the form of computational limitations. Threat detection and privacy protection using AI-powered approaches have much potential, but in contexts with limited resources, the computational demands of these methods might be too much for the current infrastructure to handle. In some deployment situations, the computer capacity needed to interpret massive volumes of data in real-time from sensors, cars, and traffic control systems could not be readily accessible. This might lead to less efficient response times, especially in highly populated areas with massive traffic data. Finally, thieves are always looking for new ways to exploit weaknesses; thus, hostile attacks are something to be worried about. Emerging attack vectors, such as zero-day vulnerabilities or complex denial-of-service assaults, can evade current protections, regardless of how advanced AI security mechanisms are. Due to the ever-changing nature of these attacks, ITS security solutions must be flexible and regularly upgraded to tackle emerging vulnerabilities.

V. CONCLUSION

GenAI can revolutionize ITS; however, additional security concerns must be addressed. Adaptive security models, real-time threat detection, and sophisticated encryption techniques are necessary to combat AI-driven cyber threats. It has tackled data privacy, intrusion detection, and secure communication, among other important issues. Addressing these issues involves striking an equilibrium between protective measures and technological advancement. Privacy-preserving techniques that enable effective data analysis without disclosing personally identifying information include data anonymization, differential privacy, and synthetic data creation. While AI-powered real-time intrusion detection systems assist in identifying and removing such threats, ellipsocurl cryptography and other cutting-edge encryption technologies aid to safeguard communication lines. The proposed AI-MA model achieves a high threat detection accuracy of 96.2%, a privacy protection score of 91.8%, a computational efficiency of 92.9%, a resilience score of 97.8%, and a network reliability ratio of 92.6% compared to other existing models. Using pre-trained AI models known to be biased and fragile is a research limitation. Shifts in processing costs and policy issues are two examples associated with real-life implementation that need further investigation. The barriers mentioned above must be removed for intelligent transport networks to be effective in further deployments.

REFERENCES

- [1] S. Chaudhary, V. Gkioulos, and S. Katsikas, "Developing metrics to assess the effectiveness of cybersecurity awareness program," *J. Cybersecurity*, vol. 8, no. 1, Jan. 2022, Art. no. tyac006.
- [2] K. Shaukat et al., "Performance comparison and current challenges of using machine learning techniques in cybersecurity," *Energies*, vol. 13, no. 10, p. 2509, May 2020.
- [3] A. Pollini et al., "Leveraging human factors in cybersecurity: An integrated methodological approach," *Cognition, Technol. Work*, vol. 24, no. 2, pp. 371–390, May 2022.
- [4] A. A. Ganin et al., "Multicriteria decision framework for cybersecurity risk assessment and management," *Risk Anal.*, vol. 40, no. 1, pp. 183–199, Jan. 2020.
- [5] M. Ahsan, K. E. Nygard, R. Gomes, M. M. Chowdhury, N. Rifat, and J. F. Connolly, "Cybersecurity threats and their mitigation approaches using machine learning—A review," *J. Cybersecurity Privacy*, vol. 2, no. 3, pp. 527–555, Jul. 2022.
- [6] M. K. Hasan, A. A. Habib, Z. Shukur, F. Ibrahim, S. Islam, and M. A. Razzaque, "Review on cyber-physical and cyber-security system in smart grid: Standards, protocols, constraints, and recommendations," *J. Netw. Comput. Appl.*, vol. 209, Jan. 2023, Art. no. 103540.
- [7] A. Torok, Z. Szalay, and B. Saghi, "New aspects of integrity levels in automotive industry-cybersecurity of automated vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 1, pp. 383–391, Jan. 2022.
- [8] S. Safavat and D. B. Rawat, "On the elliptic curve cryptography for privacy-aware secure ACO-AODV routing in intent-based Internet of Vehicles for smart cities," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 8, pp. 5050–5059, Aug. 2021.
- [9] M. Alloghani, D. Al-Jumeily, A. Hussain, J. Mustafina, T. Baker, and A. J. Aljaaf, "Implementation of machine learning and data mining to improve cybersecurity and limit vulnerabilities to cyber attacks," in *Nature-Inspired Computation in Data Mining and Machine Learning*. Cham, Switzerland: Springer, 2020, pp. 47–76.
- [10] H. Kayan, M. Nunes, O. Rana, P. Burnap, and C. Perera, "Cybersecurity of industrial cyber-physical systems: A review," *ACM Comput. Surv.*, vol. 54, no. 11s, pp. 1–35, Jan. 2022.

- [11] J. Petit and G. Le Lann, "Next generation vehicles, safety, and cybersecurity—The CMX framework," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 2, pp. 1333–1345, Feb. 2024.
- [12] A. Yaseen, "The role of machine learning in network anomaly detection for cybersecurity," *Sage Sci. Rev. Appl. Mach. Learn.*, vol. 6, no. 8, pp. 16–34, 2023.
- [13] S. Gupta Bhol, J. Mohanty, and P. Kumar Pattnaik, "Taxonomy of cyber security metrics to measure strength of cyber security," *Mater. Today, Proc.*, vol. 80, pp. 2274–2279, Jan. 2023.
- [14] M. Wazid, A. K. Das, V. Chamola, and Y. Park, "Uniting cyber security and machine learning: Advantages, challenges and future research," *ICT Exp.*, vol. 8, no. 3, pp. 313–321, Sep. 2022.
- [15] *Edge-IIoTset Cyber Security Dataset of IoT & IIoT*. Accessed: Feb. 7, 2025. [Online]. Available: <https://www.kaggle.com/datasets/mohamedamineferrag/edgeiiotset-cyber-security-dataset-of-iiot>
- [16] Y. Zhang et al., "Privacy protection for open sharing of psychiatric and behavioral research data: Ethical considerations and recommendations," *Alpha Psychiatry*, vol. 26, no. 1, p. 38759, 2025.
- [17] S. Shirvani, Y. Baseri, and A. Ghorbani, "Evaluation framework for electric vehicle security risk assessment," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 1, pp. 33–56, Jan. 2024.
- [18] H. Hufron, S. Fikri, S. Hadi, I. Shulga, and A. S. Wibowo, "Digital platform power play: Indonesian and European union law perspective," *Lex Scientia Law Rev.*, vol. 8, no. 2, pp. 707–742, Nov. 2024.
- [19] H. Khalid, S. J. Hashim, F. Hashim, W. A. M. Al-Jawher, M. A. Chaudhary, and H. H. M. Altarturi, "RAVEN: Robust anonymous vehicular end-to-end encryption and efficient mutual authentication for post-quantum intelligent transportation systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 11, pp. 17574–17586, Nov. 2024.
- [20] R. Ejjami, "The digital evolution strategies for overcoming cybersecurity and adoption challenges in French SMEs," *Int. J. Multidisciplinary Res.*, vol. 6, no. 3, May 2024.
- [21] A. Chhabra, R. Saha, and M. Conti, "PITCOR: Privacy-assured time-controlled revocation in blockchain transactions," in *Proc. 6th Int. Conf. Blockchain Comput. Appl. (BCCA)*, Nov. 2024, pp. 141–148.
- [22] M. Khodayari, M. Akbari, and P. Foroudi, "The sharing economy: A systematic literature review and research agenda," *Int. J. Consum. Stud.*, vol. 49, no. 1, Jan. 2025.
- [23] L. Caldwell, *Cybersecurity As a Human Right: A Reformulation of the Theoretical Framework of Securitization Theory*. Scottsdale, AZ, USA: Northcentral University, 2022.
- [24] C. Tran, F. Fioretto, and P. Van Hentenryck, "Differentially private and fair deep learning: A Lagrangian dual approach," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, vol. 35, no. 11, pp. 9932–9939.
- [25] E. Pacheco, "Older adults' safety and security online: A post-pandemic exploration of attitudes and behaviors," 2024, *arXiv:2403.09208*.